

ACTIVITY 5 — FACILITATOR GUIDE**Business Case and Cost-Benefit Analysis**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	90 minutes	LO1	2 - Start the Project

Purpose of this activity

By the end of this activity learners will be able to build a quantified benefits model from the Lab 03 baseline metrics.

What learners produce

business case, cba model, benefits map — submitted to artifacts/.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Laptop with internet access for the live problem-solving tools
- The course slides for the framework and worked example

Set-up

Groups of 3 to 4. Distribute one worksheet per learner and one checklist per group; each group needs one laptop for the calculations.

Scenario / briefing material

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Quantify the benefits from the Lab 03 baselines. Every benefit must trace to a metric you baselined in Lab 03. Create artifacts/05-cba-model.md.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
2	Build the cost side. Note that the total build cost equals the SGD 480,000 ceiling exactly, and that the contingency reserve sits *inside* it - the Lab 02 factor E2 consequence. You will decompose these numbers properly in Lab 13.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
3	Compute payback, NPV, ROI and BCR. Build the net cash flow table. Use a discount rate of 8%, Contoso's stated cost of capital.	Ask what the number would have to be before the decision flips. That is the exam's favourite question.
4	Stress-test against the Lab 02 downside. Lab 02 factor E1 said client L&D budgets are down 8%. Test the case	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates

	with 8% lower registration volume *and* a weaker abandonment recovery - 34% down to only 22% rather than 15%.	into every later step.
5	Choose among options. The sponsor asks you to compare three options before committing. Complete the comparison and make a recommendation.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
6	Write the business case document. Create artifacts/05-business-case.md using this structure:	Ask what the number would have to be before the decision flips. That is the exam's favourite question.
7	Build the benefits realisation map. Create artifacts/05-benefits-map.md. This is what Lab 23 will formally close against, so make it verifiable.	The trace is the learning, not the artifact. Ask a group to walk one line end to end aloud.
8	Answer the exam-style scenarios	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.

Suggested timing

Minutes	Phase
9 min	Brief the scenario, confirm the inputs and assign roles
33 min	Work the first half of the steps individually, then pool as a group
34 min	Complete the remaining steps and finalise the deliverable
14 min	Group share-back, facilitator debrief and exam-style linkage

Total: 90 minutes

Discussion & decision prompts

1. Build a quantified benefits model from the Lab 03 baseline metrics — how did your group decide, and what would change that decision?
2. Compute payback period, net present value, return on investment and benefit-cost ratio - by hand, so the exam formulas stick — how did your group decide, and what would change that decision?
3. Test the case against the Lab 02 downside scenario of an 8% fall in enrolment — how did your group decide, and what would change that decision?
4. Every benefit line traces to a Lab 03 baseline metric. A benefit with no baseline cannot be proven in Lab 23 — can every member of your group defend it?
5. Where would business case and cost-benefit analysis have saved you time or money on a real project you have worked on?

What good looks like

- Every benefit line traces to a Lab 03 baseline metric. A benefit with no baseline cannot be proven in Lab 23.
- Your discount factors decrease with time and are below 1. If any factor exceeds 1, you have inverted the formula.
- Your NPV subtracts the initial investment. Forgetting the year-0 outflow is the single most common arithmetic error here.
- Your BCR is above 1 and equals total benefit divided by total cost - not the inverse.
- Your do-nothing option has a *negative* number attached, not a blank. Do-nothing is never free.
- Your recommendation cites NPV as the deciding measure, not payback or project size.
- Your downside scenario still shows a positive NPV, and you stated that conclusion explicitly.
- In Scenario 2 you named the SGD 180,000 as sunk before discussing what to do next.

